

SIMEA – Systemic Influence Model of Economic Agency

A Proposal for a New Economic Paradigm

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August 2, 2025

Abstract

SIMEA (Systemic Influence Model of Economic Agency) reconceptualizes the economic agent as a decision-subject embedded within and guided by systems—such as algorithms, platforms, and social infrastructures—rather than as a rational decision-maker. This paradigm introduces novel constructs: the Influence Gradient (IG), Autonomy Threshold (AT), and Systemic Entropy (SE). These define how external systems shape choices, explaining phenomena like algorithmic consumption, collective irrationality, and political polarization. SIMEA proposes a welfare function that weights utility by autonomy and choice diversity, laying the foundation for a new economic framework fit for the data-driven age.

Keywords: Economic agency, systemic influence, autonomy, welfare function, decision theory

1. Introduction and Literature Gap

Classical economics defined the agent as a rational utility-maximizer (*Homo economicus*). Behavioral economics challenged this assumption through bounded rationality and cognitive biases (Simon, 1955; Kahneman & Tversky, 1979). In the 21st century, however, decision-making is increasingly shaped by opaque, persistent external systems—algorithms, platforms, and media. SIMEA shifts focus from individual choice to systemic influence, redefining economic agency for our digital age.

2. Model Structure and Mathematical Foundations

2.1 Core Decision Function

$$Decision_i = f(\vec{IG}_i, AT_i, SE) \quad (1)$$

Where:

- $\vec{IG}_i$: Aggregate Influence Gradient acting on agent i
- AT_i : Autonomy Threshold of agent i
- SE : Systemic Entropy of the environment

2.2 Influence Gradient (IG)

$$\vec{IG}_i = \sum_{k=1}^n w_k \cdot \nabla_k U_i(x) \quad (2)$$

2.3 Autonomy Threshold (AT)

$$AT_i = 1 - \sum_{k=1}^n |\vec{IG}_{ik}| \quad (3)$$

2.4 Systemic Entropy (SE)

$$SE = - \sum_{j=1}^m P_j \cdot \log P_j \quad (4)$$

2.5 Directed Utility (DU)

$$DU_i(x) = \sum_{k=1}^n \vec{IG}_{ik} \cdot B_{ik}(x) \quad (5)$$

2.6 Agent Typology

Agent Type	Description	AT Score
Homo economicus	Rational optimizer	~ 1.0
Homo psychologicus	Internally biased decisions	$0.6 - 0.8$
Homo instructus	System-guided actor	< 0.5

3. Applications and Case Studies

3.1 Algorithmic Consumption

Platforms like TikTok use opaque recommendation systems.

$\Rightarrow IG_{platform}$ high, AT low, SE low.

3.2 Financial Market Anomalies

E.g., GameStop (2021): Behavior guided by social narratives.

$\Rightarrow IG_{social} + IG_{cultural}$ dominate.

3.3 Algorithmic Voting

Targeted political content collapses alternative viewpoints.

$\Rightarrow IG_{media} + IG_{data}$ high, AT drops.

3.4 Policy Incentives

Behavior influenced not by utility but perception.

$\Rightarrow IG$ shaped by policy; impact varies with AT.

3.5 Climate and Ethical Behavior

Eco-labels and trackers guide behavior through visibility and instruction.

4. Policy Design and Welfare Economics

4.1 Classical Policy Is Insufficient

Tax and price incentives fail without altering influence structures.

4.2 Three Policy Pillars

Dimension	Goal	Policy Tools
IG Regulation	Limit manipulation	Algorithm audits, ad restrictions
AT Expansion	Increase autonomy	Media literacy, critical education
SE Preservation	Maintain diversity	Anti-monopoly, platform plurality

4.3 SIMEA Welfare Function

$$W_{SIMEA} = \sum_{i=1}^n [DU_i(x) \cdot AT_i] \cdot SE \quad (6)$$

Welfare depends on autonomy-weighted directed utility in a diverse environment.

5. Criticisms and Future Research

Criticism 1: IG and AT are hard to measure.

Response: Digital trace data enables proxy metrics.

Criticism 2: Model may seem overly deterministic.

Response: AT preserves heterogeneity and agency.

Criticism 3: Classical models seem excluded.

Response: SIMEA generalizes them; when $IG \rightarrow 0$, classical theory is recovered.

Research Agenda:

- Operationalize IG, AT, SE with empirical data
- Extend to dynamic (time-evolving) SIMEA
- Use agent-based simulations
- Apply in education, labor, health domains
- Test against behavioral models

6. Conclusion

SIMEA offers a paradigm shift: from individual choice to system-guided behavior. In data-driven societies, understanding influence becomes essential to reclaiming agency and designing effective policy. Autonomy and diversity—not just utility—are at the heart of future economics.

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